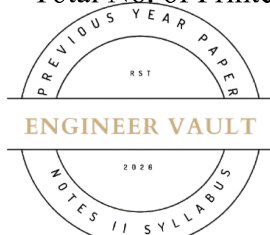


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**SUBJECT CODE NO: - RNP-8056**  
**FACULTY OF SCIENCE AND TECHNOLOGY**  
**S.E (CSE/CS/CE/IT) (NEP) Sem - IV**  
**Examination May/June-2026**  
**Probability and Statistics**

**[Time: 1:30 Hours]****[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B

- 1) Part A: Question No 1 is compulsory.
- 2) Attempt any four questions from remaining Part B: Q.2 to Q.7.

**Part A****Q.1 Solve any 5 following questions****10**

- a) What is conditional probability?
- b) State the axiomatic definition of probability.
- c) What are parameters of Poisson distribution?
- d) Define Spearman's rank correlation.
- e) What is angle between regression lines?
- f) What is Type II error?
- g) What is standard error?

**Part B****Solve any four from the following.**

- Q.2** In a class of 75 students, 17 were considered to be very intelligent, 42 as medium and the rest below average, the probability that very intelligent student fails in viva-voce examination is 0.005 the medium students failing has probability 0.05; and corresponding probability for a below average students is 0.15. If a student is known to have passed the viva-voce examination, what is the probability that he is below average? **05**
- Q.3** Explain Poisson distribution and its important properties. **05**
- Q.4** In a sample of 1000 cases, the mean of certain test is 14 and standard deviation is 2.5. assuming the distribution to be normal, find **05**
- i) How many score above 18
  - ii) How many score below 8
  - iii) How many score between 12 and 15.
- Q.5** Explain linear and non-linear regression with examples. **05**
- Q.6** A sample of size 16 has mean = 20 and standard deviation = 4. Test whether the population mean is 18 at 5% level of significance. **05**
- Q.7** Discuss Type I and Type II errors with examples. **05**

